

Scenario

Customer Support — AI Support Resolution Agent. The agent works over a synthetic ticket queue plus a small policy/FAQ knowledge base and decides whether to Auto-Resolve, Escalate, Refuse, or Ask for More Information.

Safety Principle

Replies are drafts only and are never auto-sent. Policies are quoted only from the knowledge base. If no policy is found, the agent states this and escalates rather than fabricating.

Problem Statement

Support Ticket Triage & Resolution Agent

Scenario: Customer Support — AI Support Resolution Agent.

Overview: Agent works over a synthetic ticket queue plus a small policy/FAQ knowledge base. For each ticket it decides: Auto-Resolve, Escalate, Refuse, or Ask for More Information, and drafts the reply for human approval.

Flow: Ticket In → Sentiment & Policy Check → RAG Answer Draft → LangGraph Route Decision → Confidence Re-check Loop → HITL Approval → Audit Log

Safety mapping: Replies are drafts, never auto-sent. Policies are quoted only from the KB; if no policy is found, the agent states this and escalates rather than fabricating. Refund-abuse or abusive-content requests are refused with a polite scripted response.

Lessons covered: RAG, ReAct-style reasoning, HITL approval gate, memory for customer conversation thread, Arize evaluation of answer groundedness, LangGraph conditional branching for auto-resolve / escalate / refuse / ask-clarification routes, and LangGraph loops for retrieval refinement, confidence re-check and human approval cycles.

1. Project Objective

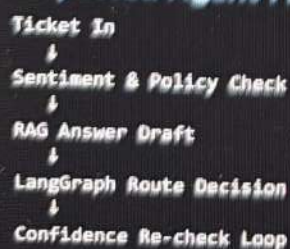
Build an AI-powered support ticket triage and resolution agent that processes customer support tickets using a synthetic ticket queue and a small policy/FAQ knowledge base.

For each incoming ticket, the agent must decide one of four actions:

- Auto-Resolve
- Escalate
- Refuse
- Ask for More Information

The agent drafts a response for human approval, but never sends the reply automatically.

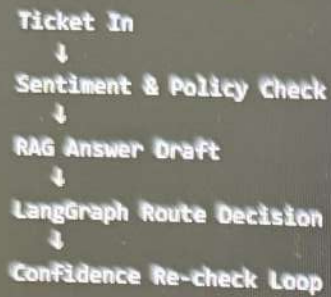
2. Expected Agent Flow



Participant-ready implementation and evaluation plan

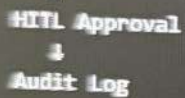
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2. Expected Agent Flow



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3. Recommended Folder Structure

```
support-ticket-agent/
├── README.md
├── requirements.txt
├── .env.example
├── config/
│   ├── app_config.yaml
│   ├── model_config.yaml
│   └── routing_rules.yaml
├── data/
│   ├── tickets/
│   │   ├── synthetic_tickets.json
│   │   └── sample_ticket_batch.json
│   ├── knowledge_base/
│   │   ├── refund_policy.md
│   │   ├── account_access_faq.md
│   │   ├── subscription_policy.md
│   │   ├── abusive_content_policy.md
│   │   └── troubleshooting_faq.md
│   └── evaluation/
│       ├── golden_dataset.json
│       └── expected_routes.json
├── src/
│   ├── main.py
│   ├── graph/
│   │   ├── support_graph.py
│   │   ├── graph_state.py
│   │   ├── nodes.py
│   │   └── edges.py
│   └── agents/
│       ├── triage_agent.py
│       └── rag_agent.py
```

```
test_ticket_flow.py
├── outputs/
│   ├── drafted_replies/
│   ├── audit_logs/
│   └── evaluation_reports/
└── docs/
    ├── architecture.md
    ├── participant_guide.md
    ├── evaluation_rubric.md
    └── demo_script.md
```

4. Key Components to Implement

A. Synthetic Ticket Queue

Participants should create or use sample tickets with ticket ID, customer ID, subject, message, conversation history, and priority.

```
{
  "ticket_id": "TCK-1001",
  "customer_id": "CUST-001",
  "subject": "Refund request for annual plan",
  "message": "I was charged for an annual plan yesterday. I forgot to cancel. Can I get a refund?",
  "conversation_history": [],
  "priority": "medium"
}
```

Recommended ticket categories:

- Refund request
- Subscription cancellation
- Login/account access issue
- Product troubleshooting
- Abusive customer message

B. Knowledge Base

Participants should build a small Markdown-based KB with policies and FAQs. The agent can only quote policies from the KB.

Refund Policy:

Customers are eligible for a refund within 7 days of billing if the product was not actively used after renewal.

Refund requests outside the 7-day window must be escalated to a human support specialist.

Repeated refund requests from the same account within 90 days should be escalated for review.

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Important Rule

If no policy is found, the agent must say that the policy could not be verified and escalate the ticket.



Normal

No Spacing

- Login/account access issue
- Product troubleshooting
- Abusive customer message

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5. HITL Approval Gate

Every drafted reply must go through a human approval step. Reviewer actions can include:

- Approve
- Reject
- Edit
- Request Regeneration
- Escalate

```
{
  "ticket_id": "TCK-1001",
  "draft_reply": "...",
  "route_decision": "AUTO_RESOLVE",
  "confidence_score": 0.91,
  "retrieved_sources": ["refund_policy.md"],
  "reviewer_action": "APPROVED",
  "reviewer_comments": "Looks good,"
}
```

Required Safety Rule

The system prepares support drafts only. It does not send responses directly to customers.

```

"retrieved_sources": ["refund_policy.md"],
"reviewer_action": "APPROVED",
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```

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6. RAG Implementation Plan

Step	Implementation Guidance
1. Load Documents	Load Markdown files from data/knowledge_base/.
2. Chunk Documents	Use 500 to 800 token chunks with 50 to 100 token overlap.
3. Create Embeddings	Create embeddings for all KB chunks.
4. Store in Vector DB	Use FAISS, Chroma, Azure AI Search, or another local vector store.
5. Retrieve Context	Retrieve top 3 to 5 relevant chunks per ticket.
6. Ground Response	Response must be based on retrieved KB content, include source references, and escalate if grounding is weak.

7. Final Submission Checklist

☐ Synthetic ticket queue included

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☐ Knowledge base files included

☐ LangGraph flow implemented

☐ RAG retrieval working

☐ Route decision working

☐ Final review and submission